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Component for attaching a cable to a housing member

Abstract

A component for attaching a cable to a housing member includes an attachment section having an attachment element and a cable clamping section integrally connected to the attachment section. The cable clamping section has a tubular shape and forms a cable tunnel. The cable clamping section has a clamping tongue that is deflectable into the cable tunnel.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of the filing date under 35 U.S.C. § 119(a)-(d) of German Patent Application No. 102021105894.5, filed on Mar. 11, 2021.

FIELD OF THE INVENTION

(2) The present invention relates to a component for attaching a cable to a housing member.

BACKGROUND

(3) A housing member and a component may be parts of a housing assembly which may be used to attach a cable to an external element, for example, to a ground connection. Such systems are used, for example, in automobiles. Previous systems, however, are difficult to assemble, are prone to assembly errors, and can be unsafe.

SUMMARY

(4) A component for attaching a cable to a housing member includes an attachment section having an attachment element and a cable clamping section integrally connected to the attachment section. The cable clamping section has a tubular shape and forms a cable tunnel. The cable clamping section has a clamping tongue that is deflectable into the cable tunnel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention will now be described by way of example with reference to the accompanying Figures, of which:

(2) FIG. 1 is an exploded perspective view of a component according to an embodiment with further elements;

(3) FIG. 2 is a sectional perspective view of the component;

- (4) FIG. 3 is a sectional side view of the component;
- (5) FIG. 4 is a perspective view of a housing member and the component connected to the further elements;
- (6) FIG. 5 is a sectional perspective view of a connection assembly in an adjustment state;
- (7) FIG. 6 is a sectional side view of a detail of the connection assembly;
- (8) FIG. 7 is a sectional perspective view of the detail of the connection assembly;
- (9) FIG. 8 is a perspective view of the connection assembly in a clamping state;
- (10) FIG. 9 is a sectional side view of a detail of the connection assembly in the clamping state; and
- (11) FIG. 10 is a sectional perspective view of the detail of the connection assembly in the clamping state.

DETAILED DESCRIPTION OF THE EMBODIMENT(S)

(12) The invention shall be explained in more detail hereafter by way of exemplary embodiments with reference to the drawings. The developments and configurations shown are each independent of one another and may be combined with one another at random, depending on necessity for the application.

(13) A component **10** for attaching a cable **90** to a housing member **50** is shown in the figures. Component **10** comprises an attachment section **20** with several attachment elements **21** and a tubular cable clamping section **30** that is connected integrally to attachment section **20** and that forms a cable tunnel **35**. Cable tunnel **35** forms a substantially circular-cylindrical receptacle for cable **90**, shown in FIG. 1, which has a circular-cylindrical configuration. The size and the shape of cable tunnel **35** are matched to the size and the shape of cable **90** so that cable tunnel **35** guides cable **90** and holds it in a stable manner. Cable **90** may be standardized, in particular according to national or international standards or according to company-specific or proprietary standards.

(14) Cable **90** may be, for example, a ground connection with which a device is connected to a ground potential or an earth potential. Cable **90** then typically comprises a single inner conductor **91** and a sheath **92** that surrounds conductor **91**, as shown in FIG. 1, and is made of insulating material, for example, elastic or plastic material.

(15) A cable lug **95** shown in FIG. 1 may be attached to one end of cable **90**, for example, by crimping cable lug **95** onto a crimping section **97** on an exposed part of conductor **91**. Cable lug **90** further comprises an opening **96** into which a pin **70**, shown in FIG. 5, may be inserted. Pin **70** may have at a ground potential or an earth potential applied.

(16) In adjustment position **201** shown in FIGS. 5, 6 and 7, cable **90** may be displaced and thereby adjusted relative to the remainder of connection assembly **200** in a cable receptacle **59** along a cable receiving direction **K** which runs parallel to an axial direction of cable tunnel **35**. The connection assembly **200** includes a housing assembly **100** described below and the cable **90** with the cable lug **95** arranged in the housing assembly **100**. In particular, the position of opening **96** may be adjusted. Once cable **90** with cable lug **95** has been inserted into a chamber **55** of a housing assembly **100**, as shown in FIG. 5, the pin **70** in the form of a dummy pin has been inserted into opening **96** along a pin receiving direction **S**. The housing assembly **100** includes the component **10** and a housing member **50** described in greater detail below.

(17) For attaching cable **90** in component **10** and affixing the position, cable clamping section **30** comprises at least one clamping tongue **31** which may be deflected into cable tunnel **35**, as shown in FIGS. 2 and 3. In the example shown, there are two oppositely disposed clamping tongues **31** in order to achieve a good distribution of forces and a good clamping effect. Clamping tongues **31** may there be deflected in and opposite to a radial direction **R** which runs perpendicular to axial direction **A**. In another embodiment, there also may be more than two clamping tongues **31** present distributed uniformly along the circumference of the clamping section **30**.

(18) Each of clamping tongues **31** comprises an inwardly directed clamping projection **32** as shown in FIGS. 2 and 3 that, when clamping tongue **31** is manipulated, presses or digs into sheath **92** of

cable **90**, as shown in FIG. **6**, and thereby prevents any movement of cable **90**. The sheath **92** is not destroyed by the clamping tongue **31**, but merely plastically deformed. A narrow configuration of the clamping projection **32**, for example, as a pin or strip, enables good plastic deformation of the cable sheath **92** with low forces at the same time. The clamping projection **32** may project into the cable tunnel **35**. In an embodiment, the clamping projection **32** is at a free end of the clamping tongue **31**.

(19) By manipulating clamping tongue **31**, connection assembly **200** then transitions to a clamping state **202** which is shown, for example, in FIGS. **8**, **9** and **10**. In clamping state **202**, the position of clamping tongue **31** is affixed by a strap-shaped or a belt-shaped clamping element **40**, presently in the form of a cable tie. A cable tie is presently to mean in particular an element with a belt-shaped toothed section and a head connected thereto in which the belt-shaped section may be affixed in a ratchet-like manner. The fixation may be repeatedly releasable or not be non-destructively releasable.

(20) In order to lock clamping element **40** against displacement along axial direction A, each clamping tongue **31** on an outer side comprises a guide groove **34**, shown in FIGS. **2**, **6**, and **7**, extending along a circumferential direction U of cable clamping section **30**. Guide grooves **34** are each defined on both sides by outwardly projecting guide strips **38** running along circumferential direction U.

(21) For simple manipulation, each clamping tongue **31** may be manipulated from the exterior. The clamping tongue **31** may project at least in sections from an outer wall of the cable clamping section **30**. This may be the case in particular in an adjustment state or a resting state, where the resting state may be a forceless state in which no forces act upon the clamping tongue **31**. The clamping tongue **31** may not project into the cable tunnel **35** in the resting state. An adjustment state may be a state in which the clamping element **40** does not yet clamp the clamping tongue **31** sufficiently and an adjustment of the cable **90** is then still possible.

(22) In order to prevent clamping projection **32** from penetrating too deeply into cable sheath **92** and damaging it, a radial cross section of cable clamping section **30** in the region of clamping tongue **31** is configured such that clamping tongue **31** deflects inwardly at most by a predefined value when clamping element **40** is attached. In the embodiment shown, this is achieved in that clamping tongue **31** is deflected inwardly at most by a predefined value when it forms a circular outer circumference with the remainder of cable clamping section **30**. When using a strap-shaped or belt-shaped clamping element **40**, it may then simply be lashed down until it is in complete abutment. The clamping tongue **31** or the clamping projection **32** may then automatically be deflected to the maximum degree.

(23) As a result, depth **37**, shown for example in FIG. **9**, to which clamping projection **32** penetrates into cable sheath **92** is effectively limited. When the strap-shaped clamping element **40** is attached and lashed to the maximum degree, it then runs in a circle along the outer circumference. This may prevent damage to the cable sheath **92** or excessive clamping force. The predefined value may correspond to a degree of the projection or be in a defined relationship therewith. For example, it may correspond to a degree of the projection in the region of a guide strip **38** minus the height of the guide strip **38** in relation to a base of the guide groove **34**.

(24) Clamping projections **32** are each beveled along axial direction A, as shown in FIG. **2**, in order to facilitate the insertion and displacement of cable **90**; the clamping tongue **31** may automatically deflect when the cable **90** is inserted. Clamping projections **32** are configured as tapering strips with an edge oriented towards cable **90**.

(25) Around each of clamping tongues **31**, cable clamping section **30** has a breakthrough **36** which frames clamping tongue **31**, as shown in FIGS. **2** and **3**. This stabilizes cable clamping section **30** and clamping tongue **31** and may limit the maximum clamping force. Clamping tongues **31** each extend approximately parallel to an axial direction A and may be deflected perpendicular thereto along radial direction R. The axial direction A may be a direction along which the cable tunnel **35**

extends through the clamping section **30**.

(26) For a good retaining effect, clamping tongues **31** are each oriented with a free end **39** towards an end **19** of component **10** on the cable side, as shown in FIG. 2.

(27) As shown in FIGS. 1, 2, and 3, attachment elements **21** on component **10** are each shaped as a latching element. They are configured as latching tabs **22** or latching hooks which interact with attachment elements **51** configured as latching projections on housing member **50**, shown in FIG. 4. The attachment elements **21**, **51** are complementary to one another. The attachment elements **21**, **51**, may also be configured differently, for example, as a screw and a corresponding hole.

(28) In the latched state, an inner stop surface **16**, which is arranged on component **10** and extends perpendicular to axial direction A, abuts against a counter-stop surface **56** on housing member **50**. For correct positioning along radial direction R, component **10** and housing member **50** each comprise approximately circular ring-shaped positioning sections **17** and **57** which are complementary to one another. The positioning sections **17**, **57** may be configured in particular in an annular, cylindrical, and/or rotationally symmetrical manner.

(29) As shown in FIG. 4, housing member **50** comprises a chamber **55** and a cable receptacle **59** connected to said chamber **55**. Chamber **55** receives pin **70** and cable lug **90** and is open along pin receiving direction S, as shown in FIG. 5. Open presently means in particular in the non-assembled state when the openings are not yet closed by other components. It has a substantially cylindrical interior. The vertical arrangement facilitates in particular an adjustment of the cable **90** relative to a pin **70** which is inserted into the chamber **55**, for example in the pin receiving direction S. Cable receptacle **59** is configured for being connected to component **10** and is open along cable receiving direction K perpendicular to pin receiving direction S. Cable receptacle **59** is formed integrally with chamber **55**. Housing member **50** is, for example, injection-molded from a plastic material.

(30) The attachment section **20** allows for reliable attachment to the housing member **50**. The cable **90** may be guided in the cable tunnel **35**, where the cable **90** may be movable when the clamping tongue **31** is not deflected into the cable tunnel **35**. This allows for the cable **90** and the connection to be adjusted.

(31) With the aid of the solution shown, it is possible to adjust and affix connection assembly **200** prior to the actual assembly such that only one pin **70** later needs to be inserted along pin receiving direction S and connection assembly **200** needs to be attached. This is achieved in particular in that the position of opening **96** is adjusted with a dummy pin prior to the actual assembly and cable **90** is thereafter affixed. After this adjustment and the removal of the dummy pin, opening **96** of cable lug **95** is centered in chamber **55** and cable **90** is clamped by way of clamping tongue **31** deflected into the cable tunnel **35**. Due to such pre-adjustment and pre-assembly when connection assembly **200** is established, the final assembly, which may take place later at a different location, is very simple and reliable.

(32) Cable lug **95** is then pressed onto pin **70** by a press-on member **60** which comprises a nut **83** and a manipulation section **82** connected thereto, shown in FIG. 5. For this purpose, pin **70** may have an external thread which is compatible with an internal thread of nut **83**. Manipulation section **82** may have a tool interface **84**, for example, with a receptacle having a hexagonal cross section, for example, in order to enable manipulation by machine. Manipulation section **82** may consist in particular of an insulating material in order to enable reliable manipulation. Nut **83** may consist of mechanically stable material, for example, metal, and may be embedded in and connected to manipulation section **82** during an injection-molding step.

(33) The connection assembly **200** comprises seals **80**, **81**, **85**, shown in FIGS. 1, 4, and 8, for sealing against dirt and liquids. The seals **80**, **81**, **85** may be configured such that they only seal when the component **10** is attached or when the press-on member **60** is manipulated.

Claims

1. A component for attaching a cable to a housing member, comprising: an attachment section having an attachment element; and a cable clamping section integrally connected to the attachment section, the cable clamping section has a tubular shape and forms a cable tunnel, the cable clamping section has a clamping tongue that is deflectable into the cable tunnel, the clamping tongue is positioned on an outer wall of the cable clamping section and is exposed to an exterior of the component in a direction perpendicular to an axial direction of the cable tunnel.
2. The component of claim 1, wherein the clamping tongue has an inwardly directed clamping projection.
3. The component of claim 1, wherein the clamping tongue has a guide groove on an outer side extending along a circumferential direction of the cable clamping section.
4. The component of claim 1, wherein the clamping tongue is manipulated from the exterior.
5. The component of claim 1, wherein a radial cross-section of the cable clamping section in a region of the clamping tongue is configured such that the clamping tongue deflects inwardly at most by a predefined value when a clamping element is attached.
6. The component of claim 5, wherein the clamping tongue is deflected inwardly at most by the predefined value when it forms a circular outer circumference with a remainder of the cable clamping section.
7. The component of claim 1, wherein the cable clamping section has a breakthrough that frames the clamping tongue.
8. The component of claim 1, wherein the clamping tongue runs parallel to the axial direction.
9. The component of claim 1, wherein a free end of the clamping tongue is oriented toward an end of the component.
10. The component of claim 1, wherein the cable tunnel has a cylindrical cross-section.
11. The component of claim 1, wherein the attachment element is a latching element.
12. The component of claim 1, further comprising an inner stop surface extending perpendicular to the axial direction.
13. A housing assembly, comprising: a component including an attachment section having an attachment element and a cable clamping section integrally connected to the attachment section, the cable clamping section has a tubular shape and forms a cable tunnel, the cable clamping section has a clamping tongue that is deflectable into the cable tunnel, the clamping tongue is positioned on an outer wall of the cable clamping section and is exposed to an exterior of the component in a direction perpendicular to an axial direction of the cable tunnel; and a housing member having a chamber and a cable receptacle connected to the chamber, the cable receptacle is connected to the component.
14. A connection assembly, comprising: a housing assembly including: a component including an attachment section having an attachment element and a cable clamping section integrally connected to the attachment section, the cable clamping section has a tubular shape and forms a cable tunnel, the cable clamping section has a clamping tongue that is deflectable into the cable tunnel, the clamping tongue is positioned on an outer wall of the cable clamping section and is exposed to an exterior of the component in a direction perpendicular to an axial direction of the cable tunnel; and a housing member having a chamber and a cable receptacle connected to the chamber, the cable receptacle is connected to the component; and a cable, an end of the cable having a cable lug is arranged in the housing assembly.
15. The connection assembly of claim 14, wherein the cable lug has an opening centered in the chamber.
16. The connection assembly of claim 14, further comprising a clamping element clamping the cable and deflecting the clamping tongue.
17. The connection assembly of claim 16, wherein the clamping element is separate from the component and extends circumferentially around the cable clamping section.

18. The connection assembly of claim 17, wherein the clamping element is a cable tie that is affixed in a ratchet-like manner.

19. The component of claim 1, wherein the cable clamping section is positioned adjacent to the attachment section and extends from the attachment section along the axial direction.

20. The component of claim 19, wherein the cable clamping section is monolithically formed in a single piece with the attachment section.
